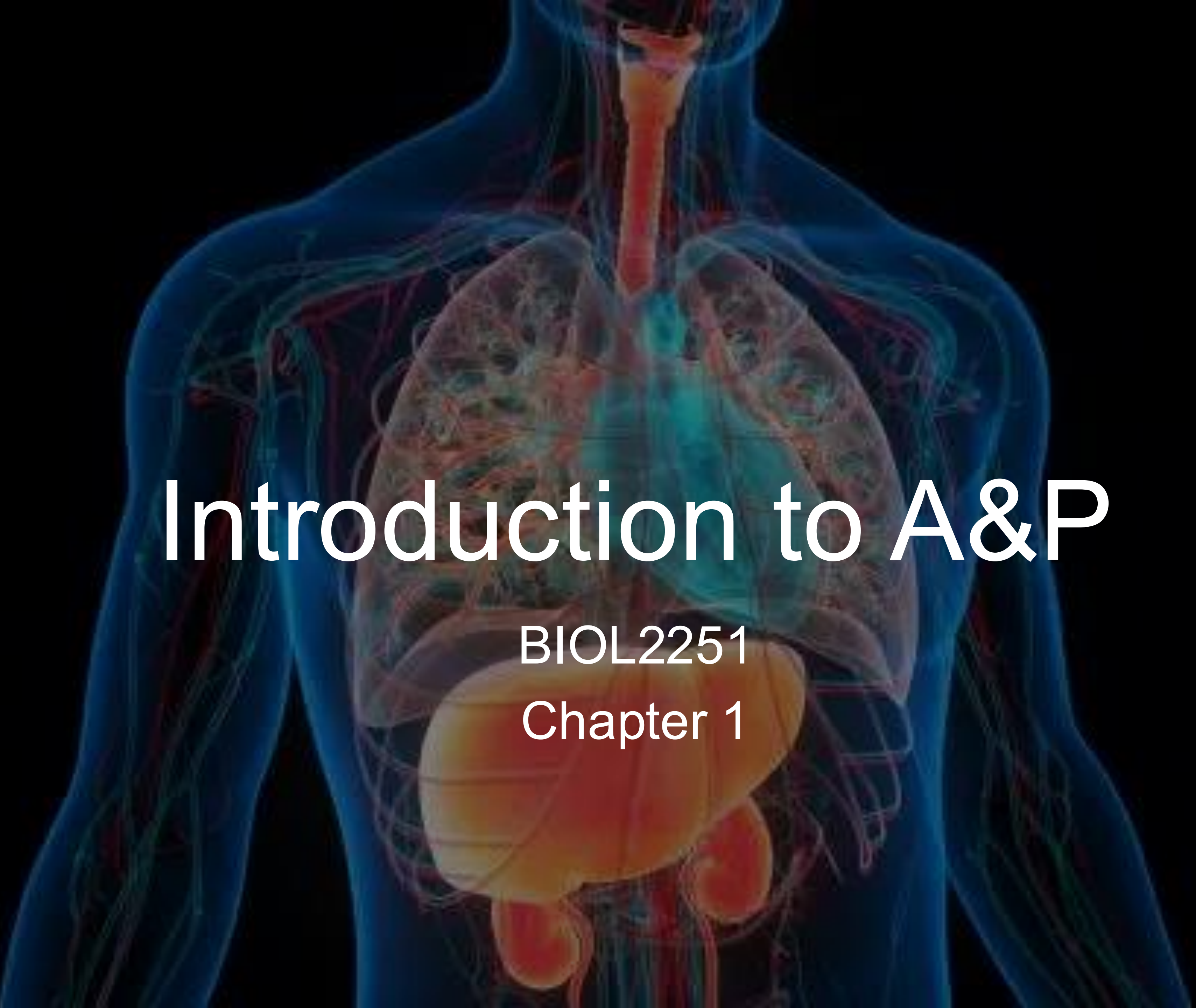


Introduction to A&P

BIOL2251

Chapter 1

No notes! Just you and your brain

Lecture Quiz

Test what you know!

10 questions (MC + **1 bonus**)

Quiz automatically submits at 1:00 PM

Password = **homeostasis**

All electronic devices away after the quiz!

Check What We Know…

Connecting Anatomy and Physiology

Connecting Anatomy and Physiology

- **Instructions**
  - On your own, **describe how you define anatomy and physiology**
  - Then in small groups, **describe the relationship between anatomy and physiology using a specific organ system**
- **Bloom’s taxonomy = Understand, Analyze**

Connecting Anatomy and Physiology

- **Anatomy** = study of body structures (form)
- **Physiology** = study of body functions (function)
- **Relationship** = structure determines function + function influences structure
- **Example:** Heart has four chambers (anatomy) → allows separation of oxygenated/deoxygenated blood for pumping (physiology)

Medical Terminology Speed Round

Medical Terminology

- **Instructions**
  - Work in small groups (4-6 people)
  - I will call out a medical term with 2-3 root words
  - You have 30 seconds to **write the definition by breaking down the root words**
  - **Points for correct definition - most points wins!**
- **Bloom's taxonomy = Remember, Apply**

Medical Terminology

- **Gastroenterology** = gastr(o) + enter(o) + logy = study of stomach and intestines
- **Arthroscopy** = arthr(o) + scopy = visual examination of a joint
- **Cardiomyopathy** = cardi(o) + my(o) + path(o) = disease of heart muscle
- **Osteoblast** = oste(o) + blast = bone-forming cell
- **Nephrology** = nephr(o) + logy = study of kidneys
- **Dermatitis** = derm + itis = inflammation of skin
- **Hyperglycemia** = hyper + glyc + emia = high blood sugar
- **Endocardium** = endo + cardi(o) = inner lining of heart

Nested Levels of Organization

Medical Terminology

- **Instructions**
  - Work in small groups (4-6 people) with whiteboards
  - **Draw a pyramid/hierarchy of the levels of organization**
    - **Work from simplest to most complex**
  - **Provide a specific example (e.g., biceps brachii muscle) and trace it through the levels**
  - I will call on 1-2 groups to share
- **Bloom's taxonomy = Understand, Analyze**

Medical Terminology

- **Chemical:** Carbon, hydrogen, oxygen, nitrogen atoms; actin and myosin protein molecules
- **Cellular:** Muscle cells (fibers)
- **Tissue:** Skeletal muscle tissue
- **Organ:** Biceps brachii muscle
- **Organ System:** Muscular system
- **Organism:** Human (whole body)

Simon Says: Anatomical Directions

Anatomical Directions

- **Instructions**
  - I will call out positions or directions, and you must move your body accordingly
- **Bloom's taxonomy = Apply**

Anatomical Directions

- “Simon says: Assume the standard anatomical position.”
  - Students should be in anatomical position
- "Simon says: Point to a region distal to your knee.”
  - Students should point to their ankle or toes.
- "Simon says: Place your hand on the lateral aspect of your thigh.”
  - Students should touch the outside of their thigh.
- "Simon says: Move your hand medially from your shoulder.”
  - Students should move their hand toward their sternum.
- "Now, assume a supine position.”
  - Students should mimic lying on their backs/palms up.
    - Catch: I didn't say Simon Says!

Directional Term Translation

Directional Term Translation

- **Instructions**
  - Work in pairs
  - I will project a list of 10 anatomical relationships using common terms
  - Your job is to rewrite the phrases using the correct directional terms
- **Bloom's taxonomy = Apply**

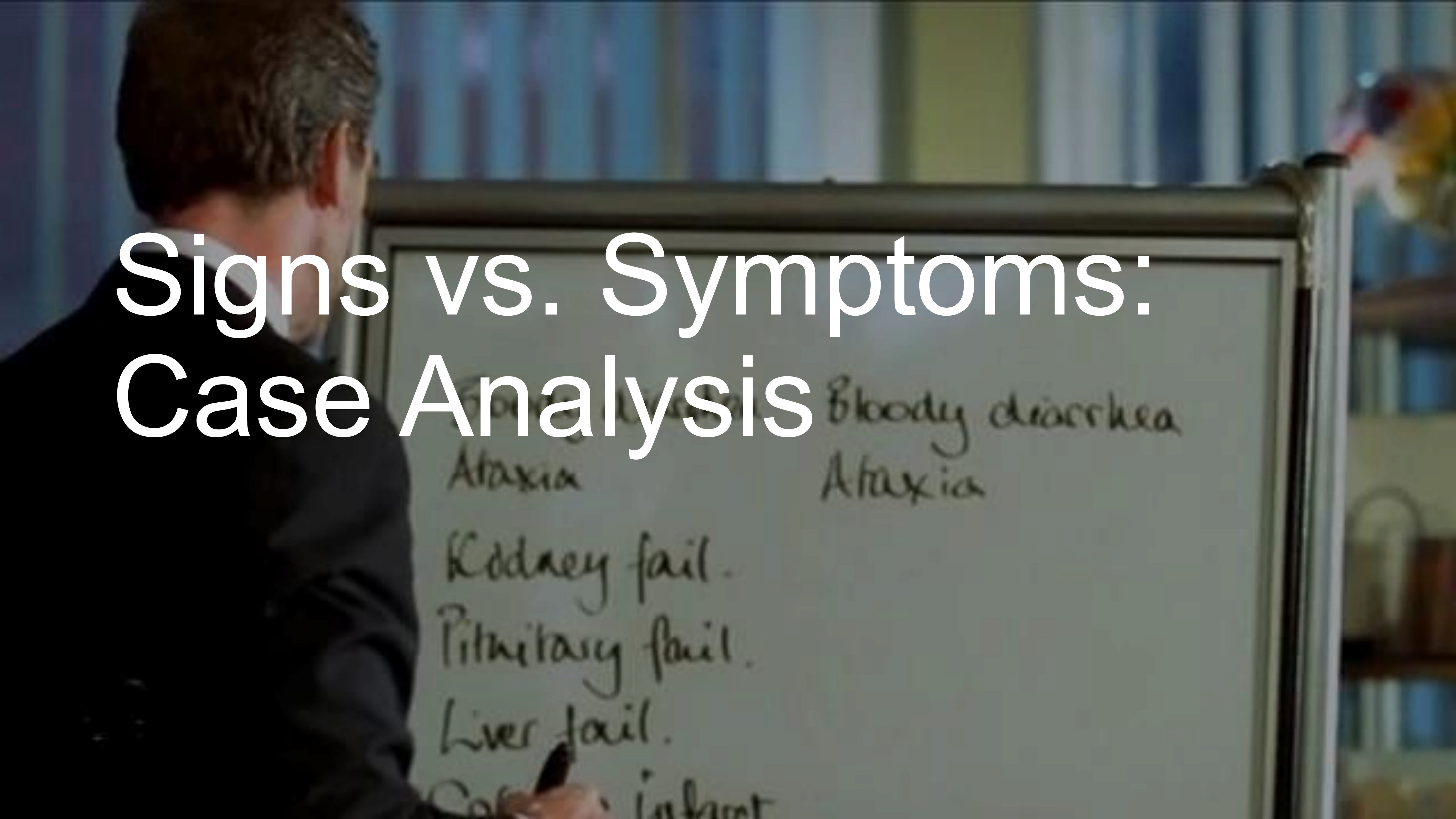
Directional Term Translation

- The nose is **between** to the ears
- The heart is **above** to the stomach
- The skin is **above** to the muscles
- The elbow is **closer** to the wrist
- The spine is **behind** to the heart
- The lungs are **to the side** of the heart
- The knee is **further away** to the hip
- The sternum is **in front** of the spine
- The stomach is **below** the lungs
- The muscles are **further inside** compared to the skin

Directional Term Translation

- The nose is **medial** to the ears
- The heart is **superior** to the stomach
- The skin is **superficial** to the muscles
- The elbow is **proximal** to the wrist
- The spine is **posterior** (dorsal) to the heart
- The lungs are **lateral** to the heart
- The knee is **distal** to the hip
- The sternum is **anterior** (ventral) to the spine
- The stomach is **inferior** to the lungs
- The muscles are **deep** to the skin

Signs vs. Symptoms: Case Analysis



Signs vs. Symptoms

- **Instructions**
  - **Consider the following case on your own**
    - A patient arrives at the clinic complaining of feeling feverish and experiencing sharp chest pain when breathing.
    - The physician measures a temperature of 102°F, observes rapid breathing, and hears crackling sounds in the lungs with a stethoscope.
  - **Classify pieces of information into signs and symptoms**, then discuss with a partner
  - As a class, **we will share ideas and discuss the pros and cons of signs vs. symptoms**
- **Bloom's taxonomy = Analyze, Evaluate**

Signs vs. Symptoms

- **Symptoms (subjective):** Feeling feverish; sharp chest pain when breathing
- **Signs (objective):** Temperature of 102°F; rapid breathing (observed); crackling lung sounds

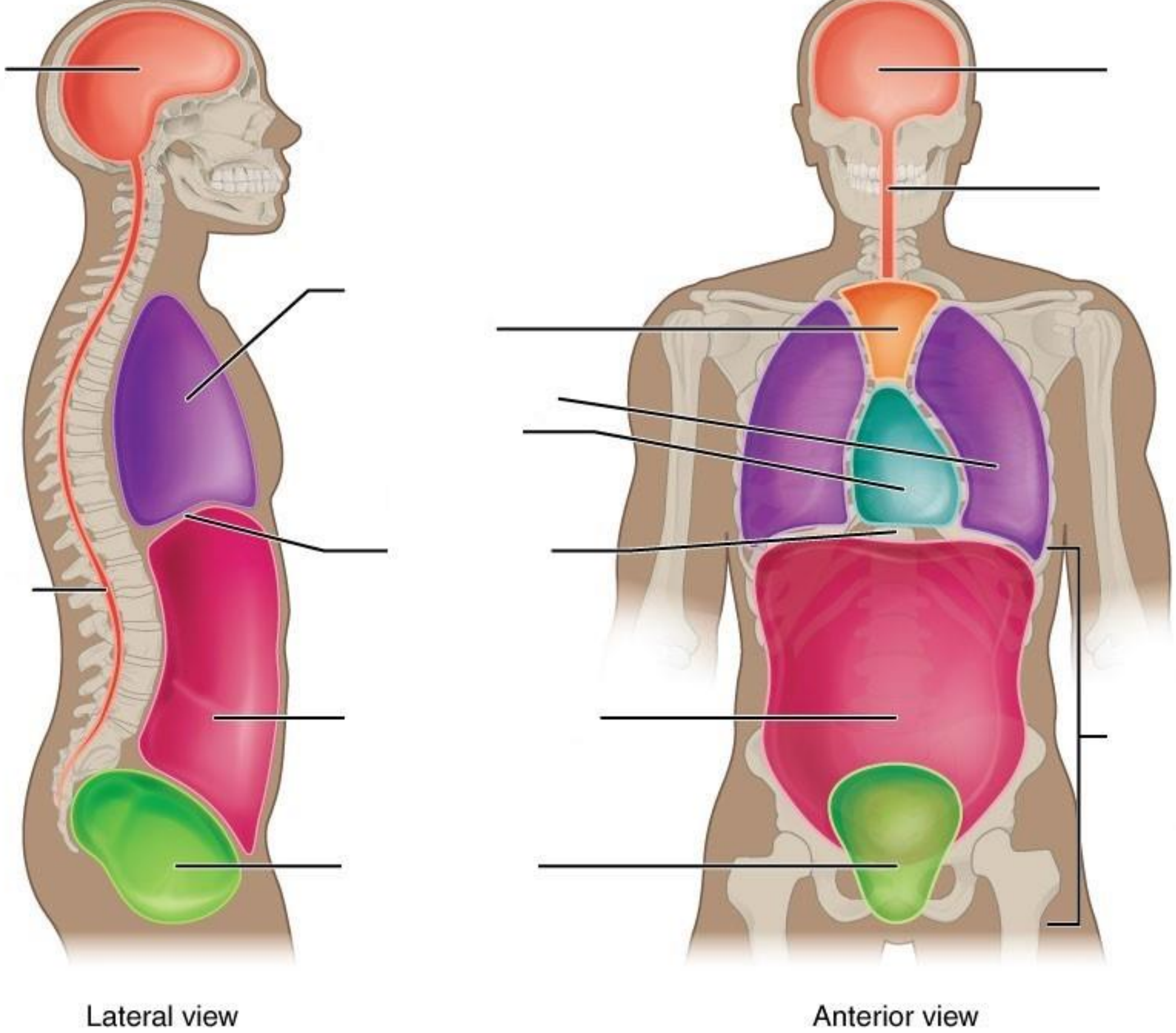
Pros/Cons:

- Signs: More objective/measurable (pro), but may miss patient experience (con); require medical equipment/expertise (con)
- Symptoms: Capture patient experience (pro), guide initial assessment (pro), but subjective/variable (con); patients may not report accurately (con)

Abdominopelvic Mapping Challenge

Abdominopelvic Mapping

- **Instructions**
  - Work on your own
  - Draw a basic outline of a human torso
    - **Divide it into 4 quadrants and label**
  - On a second drawing, **divide the torso into 9 regions and label**
  - **Identify the correct quadrant and region for the following organs:**
    - Liver, stomach, appendix, gallbladder
- **Bloom’s taxonomy = Remember, Apply**


Lateral view
Anterior view

Abdominopelvic Mapping

- **4 Quadrants:**
  - RUQ (Right Upper Quadrant)
  - LUQ (Left Upper Quadrant)
  - RLQ (Right Lower Quadrant)
  - LLQ (Left Lower Quadrant)
- **9 Regions (top to bottom, left to right):**
  - Right hypochondriac, Epigastric, Left hypochondriac
  - Right lumbar, Umbilical, Left lumbar
  - Right inguinal, Hypogastric, Left inguinal
- **Organ Locations:**
  - Liver: RUQ / Right hypochondriac + epigastric
  - Stomach: LUQ / Epigastric + left hypochondriac
  - Appendix: RLQ / Right inguinal
  - Gallbladder: RUQ / Right hypochondriac

Plane Slicing Visualization

Plane Slicing Visualization

- **Instructions**
  - I will be your reference subject
  - **I will call out a plane (sagittal, frontal, transverse) and a volunteer must indicate how they would ‘slice’ the body**
  - Follow-up about **which organs/structure would be visible in that cross-section**
- **Bloom’s taxonomy = Understand, Apply**

Plane Slicing Visualization

- **Sagittal Plane:** Vertical division into left and right portions (midsagittal = equal halves)
  - View: Nose, brain, spinal cord, heart (side view of internal organs)
- **Frontal (Coronal) Plane:** Vertical division into anterior and posterior portions
  - View: Face, front of brain, front of lungs and heart
- **Transverse (Horizontal) Plane:** Horizontal division into superior and inferior portions
  - View: Cross-section showing organ shapes (e.g., circular lungs, round vertebrae)

Organ System ‘Speed Dating’

Organ System ‘Speed Dating’

- **Instructions**
  - We will work in rotating pairs
  - You will be given an organ system
  - **Take 1 minute to share with a partner: (1) organ system name, (2) 2-3 major organs, and (3) a primary function**
    - **After 1 minute, your partner shares**
  - We will go through 6 full rotations, after which you will **write as many systems with as much detail as you can recall**
- **Bloom’s taxonomy = Remember, Understand**

Organ System ‘Speed Dating’

- **Integumentary:** Skin, hair, nails | Protection, temperature regulation
- **Skeletal:** Bones | Support, protection, movement, mineral storage
- **Muscular:** Skeletal, cardiac, smooth muscles | Movement, heat production
- **Nervous:** Brain, spinal cord, spinal nerves | Control/coordination via electrical signals
- **Endocrine:** Pituitary, thyroid, adrenals, pancreas | Control/coordination via hormones
- **Cardiovascular:** Heart, blood vessels, blood | Transport nutrients, gases, wastes
- **Lymphatic/Immune:** Lymph nodes, spleen, thymus | Immunity, fluid balance
- **Respiratory:** Lungs, trachea | Gas exchange (O_2/CO_2)
- **Digestive:** Stomach, intestines, liver, pancreas | Nutrient breakdown and absorption
- **Urinary:** Kidneys, bladder, urters | Waste removal, fluid/electrolyte balance
- **Reproductive:** Ovaries/testes, uterus/penis | Production of offspring

Body Cavity Jigsaw

Body Cavity Jigsaw

- **Instructions**
  - Work in small groups (4-6 people)
  - **Your group will be assigned either the thoracic or abdominopelvic cavity**
  - **Take 5 minutes to become ‘experts’ on your cavity**
    - **Functions, organs, membranes**
  - We will regather into mixed groups where each cavity is represented
    - **Groups will teach each other about their cavity**
  - We will end with a whole-class debrief
- **Bloom’s taxonomy = Remember, Understand, Analyze**

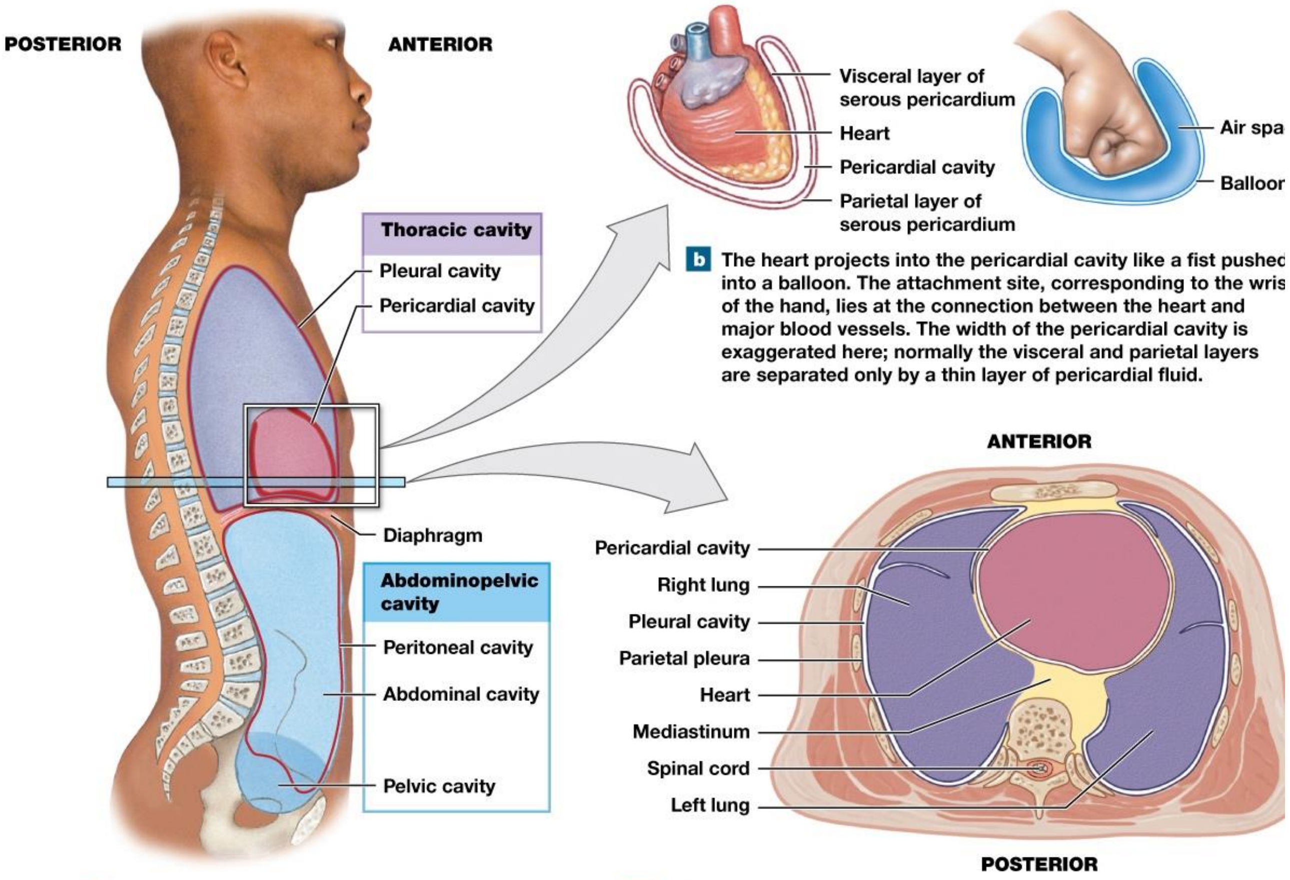
Body Cavity Jigsaw

- **Thoracic Cavity:**
  - **Subdivisions**: Pleural cavities (2), mediastinum (contains pericardial cavity)
  - **Function**: Protection, allows organ movement
  - **Organs**: Lungs (pleural), heart (pericardial), esophagus, trachea (mediastinum)
  - **Membranes**:
    - Pleura: Parietal pleura (lines cavity wall), visceral pleura (covers lung surface)
    - Pericardium: Parietal pericardium (outer), visceral pericardium/epicardium (on heart)

- **Abdominopelvic Cavity:**
  - **Subdivisions**: Abdominal cavity, pelvic cavity
  - **Function**: Protection, allows organ movement
  - **Organs**: Stomach, liver, intestines, kidneys, bladder, reproductive organs
  - **Membranes**: Peritoneum: Parietal peritoneum (lines wall), visceral peritoneum (covers organs)

- **General Cavity Function**: Protect organs, allow expansion/movement, provide lubrication (reduce friction)

Body Cavity Jigsaw



a A lateral view showing the body cavities of the trunk. The muscular diaphragm subdivides them into a superior thoracic cavity and an inferior abdominopelvic cavity. Three of the four adult true body cavities are shown and outlined in red; only one of the two pleural cavities can be shown in a sagittal section.

b The heart projects into the pericardial cavity like a fist pushed into a balloon. The attachment site, corresponding to the wris of the hand, lies at the connection between the heart and major blood vessels. The width of the pericardial cavity is exaggerated here; normally the visceral and parietal layers are separated only by a thin layer of pericardial fluid.

c A transverse section through the thoracic cavity, showing the centra location of the pericardial cavity. The mediastinum and pericardial cavity lie between the two pleural cavities. Note that this transverse or cross-sectional view is oriented as though you were standing at the feet of a supine person and looking toward that person's head. This inferior view of a transverse section is the standard presentatio for clinical images. Unless otherwise noted, transverse or cross-sectional views in this text use this same orientation (*see Clinical Note: Diagnostic Imaging Techniques*).

Medical Imaging Match-Up

Medical Imaging Match-Up

- **Instructions**
  - Work in small groups (4-6 people)
  - **You will analyze 5 brief clinical scenarios**
    - **Choose the most appropriate imaging technique for each scenario**
    - Justify your choice based on the **strengths and limitations of each method**
- **Bloom's taxonomy = Remember, Understand, Analyze**

Medical Imaging Match-Up

Scenarios

- Suspected broken arm
- Monitoring fetal development
- Suspected stroke/brain bleed
- Torn knee ligament (ACL)
- Detecting metastatic cancer

Imaging Techniques

- X-ray
- CT scab
- MRI
- PET scan
- Ultrasound

Medical Imaging Match-Up

- **Suspected broken arm** → X-ray (fast, cheap, excellent for bone)
- **Monitoring fetal development** → Ultrasound (safe, no radiation, real-time)
- **Suspected stroke/brain bleed** → CT (fast, good emergency imaging for brain)
- **Torn knee ligament (ACL)** → MRI (excellent soft tissue detail)
- **Detecting metastatic cancer** → PET scan (shows metabolic activity of cancer cells)

Feedback Loop Sorting

Feedback Loop Sorting

- **Instructions**
  - Think on your own at first
  - I will provide physiological scenarios, and **you will classify each scenario as negative or positive feedbacks**
  - After classify each scenario on your own, **discuss your answers and reasoning in small groups**
  - We will end with a whole-class debrief
- **Bloom's taxonomy = Understand, Analyze, Evaluate**

Feedback Loop Sorting

- **Body temperature regulation**
  - Too hot → sweating/vasodilation → cooling
- **Blood glucose regulation**
  - High glucose → insulin release → glucose uptake → lower glucose
- **Childbirth**
  - Contractions → oxytocin release → stronger contractions → more oxytocin
    - Stops when baby delivered
- **Blood clotting**
  - Injury → platelet activation → more platelets activated → clot forms
    - Stops when vessel sealed)
- **Blood pressure regulation**
  - High BP → baroreceptors detect → heart rate decreases → BP drops
- **Breathing rate**
  - High CO_2 → increased breathing → CO_2 expelled → CO_2 normalizes
- **Breastfeeding**
  - Suckling → prolactin release → milk production → more suckling

Feedback Loop Sorting

- **Body temperature regulation = Negative**
  - Too hot → sweating/vasodilation → cooling
- **Blood glucose regulation = Negative**
  - High glucose → insulin release → glucose uptake → lower glucose
- **Childbirth = Positive**
  - Contractions → oxytocin release → stronger contractions → more oxytocin
    - Stops when baby delivered
- **Blood clotting = Positive**
  - Injury → platelet activation → more platelets activated → clot forms
    - Stops when vessel sealed)
- **Blood pressure regulation = Negative**
  - High BP → baroreceptors detect → heart rate decreases → BP drops
- **Breathing rate = Negative**
  - High CO_2 → increased breathing → CO_2 expelled → CO_2 normalizes
- **Breastfeeding = Positive**
  - Suckling → prolactin release → milk production → more suckling

Homeostatic Disruption Detective

Homeostatic Disruption Detective

- **Instructions**
  - Work in small groups (4-6 people)
  - **We will work through 2 complex, clinical scenarios**
  - In your groups, you will:
    - **Identify the disrupted variable**
    - **Predict the body's response**
    - **Identify if it is a negative or positive feedback**
- **Bloom's taxonomy = Understand, Analyze, Evaluate**

Homeostatic Disruption Detective

Scenario 1

- **A construction worker suffers severe bleeding from an injury**
- Platelets adhere to the injured site and release chemicals that attract more platelets
- Platelets continue attracting and adhering until a clot is formed

Scenario 2

- **A person eats a meal high in fats and carbohydrates, and blood glucose levels rise**
- Pancreas releases insulin, promoting glucose uptake by the cells and reducing blood glucose levels
- **Person has Type 2 diabetes (uptake of glucose is impaired)**
- Uptake continue until normal glucose levels are restored

Homeostatic Disruption Detective

Scenario 1

- **Disrupted variable:** Decreased blood volume/pressure
- **Body's response:**
  - Negative feedback: Vasoconstriction, increased heart rate, ADH release
  - Positive feedback: Blood clotting cascade activated
- **Type:** BOTH mechanisms working simultaneously (students should at least identify positive feedback)
- If compensation fails: Hemorrhagic shock, death

Scenario 2

- **Disrupted variable:** Elevated blood glucose
- **Body's response:**
  - Negative feedback
  - Excess glucose excreted in urine, increased urination, thirst
  - Body breaks down fats and proteins for energy
    - If signaling from insulin is impaired
- **Type:** Negative feedback mechanism
- If compensation fails: Diabetic ketoacidosis, coma, death